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## INDUSTRY EXPERIENCE

**Software Engineer 2** **Microsoft - Xbox Personalization** **Oct 2024 - Present**

- Architected and built a **machine learning** powered **game recommendation** system for **Xbox**, combining **personalized** and **popularity-based candidate retrieval** with **real-time player signals**, **eligibility filtering**, and **deep-learning ranking** to deliver relevant **game recommendations**.
- Developed the **Gaming Copilot**, an artificial intelligence assistant for the Xbox App, which saw adoption by over **70%** of users within a day of its release, demonstrating strong engagement and demand.
- Built the AI-powered **Highlight Reels** system for the **Xbox ROG Ally X**, enabling real-time detection and automatic capture of key gameplay moments.

**Software Engineer 2** **XBox - Trust and Safety** **Aug 2022 - Sept 2024**

- Architected and deployed **Small Language Models (SLM)** to enable proactive content moderation, increasing detection accuracy by **65%** across Xbox platforms.
- Enhanced **reactive moderation workflows**, streamlining internal tools to boost efficiency by **18%**, reducing human moderation workload.
- Implemented **proactive moderation** strategies, mitigating **spam and fraud threats** by nearly **40%**, safeguarding Xbox users.

**Software Engineer** **Neurotrack Technologies Inc.** **May 2021 - July 2022**

- Engineered **non-invasive cognitive tests** leveraging **AWS Microservices**, improving early **Alzheimer's** and **dementia detection** accuracy, and identifying onset up to **3 years earlier** than traditional methods.
- Pioneered a **coaching service** to enhance cognitive health, resulting in a **15% improvement** in client cognitive scores, positively impacting **100+ clients** within the first **3 months**.
- Optimized **backend infrastructure**, improving **scalability** and reducing **server response time** by **25%**, ensuring **seamless user experiences.**

**Research Engineer (Mitacs Accelerate)** **Ethically Aligned AI** **Mar 2021 – May 2021**

- Analyzed literature across different domains in order to produce **proof-of-concept** for **validating ethics** in **AI**.
- Iterated **proof-of-concept** upon **feedback** from a diverse class of **businesses** to identify specific business needs for building **fair machine learning systems.**

**Software Engineer** **Famwork** **Dec 2020 – May 2021**

- Leveraged **data science, visualization** and **statistics** for **integrating machine learning models** into **electronic health record systems** and built a **chatbot** for the patient portal.
- **Spearheaded** the **automation of the telehealth system** for the minimum viable product and improved accuracy of the system to **forecast diseases** the patient was susceptible to acquire, by **80%**.

## EDUCATION/DEGREE

**M.Sc. Computing Science**, University of Alberta, Edmonton.
**B.E. in Computer Science**, Anna University, Chennai, India

## SELECTED PROJECTS

- **Noise Eliminator in Active Learning (NEAL): Algorithms** to produce **highly accurate machine learning predictions** with a very **low amount** of **training instances. Improved** over **previous systems** by **30%**. **Python.**
- **Hybrid_UI - Intelligent User Interfaces:** An **intelligent system** to improve **user interface** experience on websites for **geriatric** and the **visually impaired population.** Performed **data analysis and** tested using **hypothesis testing (A/B testing)** and conducted **experiments** on a control population of **30 people**. **Python**
- **Expense Tracker: Web app** that parses **multi-bank statement PDFs** into **categorized spending insights** via custom parsers, an auto-categorization engine, and interactive dashboards. **Python**, **Streamlit**, **Supabase** (Postgres), **OAuth**.
- **Depression Detection using Linguistic Metadata:** A **machine learning** based system that detects the **early onset of depression** in individuals. Trained using a privatized **RSDD dataset**. **Python**.

## SKILLS

| | |
|---|---|
| **Programming languages:** | Python, C#, Java, C++, HTML, CSS, JavaScript, PHP, TypeScript, React |
| **Libraries:** | NumPy, Pandas, Matplotlib, Scikit-Learn, Keras, NLTK, Gensim, spaCy, TensorFlow, PyTorch |
| **Databases:** | SQL, Postgres, CosmosDB, DynamoDB, Redis |
| **Programming Tools:** | Jupyter Notebook, Google Colab, Github, Excel, Git |
| **Platforms:** | Microsoft Windows, Ubuntu Linux, macOS |
| **Cloud** : | Amazon AWS, Microsoft Azure |
| **Tools:** | Docker, Kubernetes, Redis, Cursor, Cline, Github Copilot |
| **API:** | REST, GraphQL, Websocket |